# Python Conditional Statements

---

## 1️⃣ What are Conditional Statements?

- Conditional statements allow a program to **make decisions** based on certain conditions.
- They **execute a block of code only if a condition is True**.
- Common use: **check user input, compare values, control program flow**.

---

## 2️⃣ The `if` Statement

### Syntax:

```
if condition:
    # code to execute if condition is True
```

### Example:

```
num = int(input("Enter a number: "))
if num % 2 == 0:
    print("The number is even")
```

✅ **Explanation:**

- `if num % 2 == 0:` → checks if number is divisible by 2
- If condition is **True**, it executes the print statement

- If condition is **False**, it does nothing

---

## 3️⃣ The `if-else` Statement

### Syntax:

```
if condition:
    # code if condition is True
else:
    # code if condition is False
```

### Example:

```
num = int(input("Enter a number: "))
if num % 2 == 0:
    print("Even")
else:
    print("Odd")
```

✅ **Explanation:**

- Executes the first block if condition is True
- Executes the `else` block if condition is False

---

## 4️⃣ The `if-elif-else` Statement

- Used when you have **multiple conditions** to check.
- `elif` stands for **“else if”**.

### Syntax:

```
if condition1:
    # code if condition1 is True
elif condition2:
    # code if condition2 is True
else:
    # code if none of the above are True
```

### Example 1 — Check positive, negative, or zero:

```
num = int(input("Enter a number: "))


if num > 0:
    print("Positive")
elif num < 0:
    print("Negative")
else:
    print("Zero")
```

### Example 2 — Grade Finder:

```
marks = int(input("Enter marks: "))
```

```
if marks >= 90:
    print("Grade A")
elif marks >= 75:
    print("Grade B")
elif marks >= 60:
    print("Grade C")
else:
    print("Fail")
```

✅ **Explanation:**

- The program checks conditions **top-down**
- Executes the **first condition that is True**
- If none match, executes the `else` block

---

## 5️⃣ Key Points

- Only **one block executes** in an `if-elif-else` chain
- Conditions are **checked in order**
- Use **comparison operators** (`>`, `<`, `==`, `>=`, `<=`, `!=`)
- Logical operators can combine conditions: `and`, `or`, `not`